

# Antenna Datasheet

**Product OC:** YECT006W1EC
**Version:** 1.0
**Date:** 2026-05-25
**Status:** Released

**Product Name:** 4G Terminal Mount Rubber Monopole External Antenna
**Key Features:**
Frequency Band: 698–960 MHz, 1710–2690 MHz
Dimensions: Φ 9.4 mm × 44.6 mm
Efficiency: Up to 79.8 % (EVB)
RoHS Compliant

# Overview

YECT006W1EC is a 4G rubber antenna measuring Φ 9.4 mm × 44.6 mm. This ultra-wide-band 4G antenna provides broad coverage from 698–960 MHz, 1710–2690 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is terminated with SMA Male connector. This low profile, terminal mount omni-directional antenna, ideal for applications where the antenna is required to be discrete, is easy to install with maximum durability assured thanks to its TPE enclosure. It is compatible with Quectel's 4G Series modules.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. YECT006W1EC is designed as a monopole antenna, which needs to be mounted on a ground plane to offer high efficiency in all working bands. It is a perfect antenna product for customers that desire highest performance. This high-efficiency, high-gain omni-directional antenna is ideally suited for Zigbee, Bluetooth, IoT Sensors, public safety and security, point of sales terminals, smart home automation, robotics / autonomous.

- **Typical applications include:**

✓ Zigbee
✓ Bluetooth
✓ IoT Sensors
✓ Public Safety and Security
✓ Point of Sales Terminals
✓ Smart Home Automation
✓ Robotics / Autonomous

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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# 1 Specification

**Test Condition: Free Space & On 130 mm × 130 mm EVB**

## 1.1. Electrical

| Electrical | |
|---|---|
| **Frequency Range** | 698–960 MHz, 1710–2690 MHz |
| **Impedance** | 50 Ω |
| **Polarization** | Linear |
| **Radiation Pattern** | Omni-directional |
| **Antenna Type** | Monopole |

| Electrical – Detail | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Band / SPEC | | B12 /B13 /B28 | B5 /B8 /B26 | n74 /n75 /n76 | B1 /B2 /B3 | B40 | Wi-Fi 2G | B38 /B41 | B42 /B48 /n77 | n79 | Wi-Fi 5G |
| | Freq. (MHz) | 700–810 | 820–960 | 1420–1520 | 1700–2170 | 2300–2400 | 2400–2500 | 2500–2690 | 3300–4200 | 4400–5000 | 5150–5850 |
| Max. VSWR | FS | 9.5 | 5.6 | - | 13.5 | 2.2 | 4.1 | 8.8 | - | - | - |
| | EVB | 11.4 | 3.0 | - | 3.0 | 1.7 | 1.7 | 3.6 | - | - | - |
| Max. Return Loss (dB) | FS | -1.8 | -3.1 | - | -1.3 | -8.4 | -4.3 | -2.0 | - | - | - |
| | EVB | -1.5 | -6.1 | - | -5.9 | -11.4 | -11.7 | -4.9 | - | - | - |
| AVG Eff. (%) | FS | 14.1 | 24.1 | - | 29.9 | 45.3 | 39.7 | 24.4 | - | - | - |
| | EVB | 37.0 | 63.6 | - | 66.4 | 76.5 | 73.3 | 61.4 | - | - | - |
| AVG AVG Gain (dB) | FS | -8.6 | -6.2 | - | -6.1 | -3.5 | -4.1 | -6.3 | - | - | - |
| | EVB | -4.4 | -2.0 | - | -1.8 | -1.2 | -1.4 | -2.2 | - | - | - |
| Max. Peak Gain (dBi) | FS | -1.9 (810) | 1.5 (950) | - | 2.8 (2010) | 2.5 (2310) | 2.5 (2460) | 1.3 (2520) | - | - | - |
| | EVB | 0.9 (810) | 3.3 (930) | - | 3.0 (2090) | 3.9 (2390) | 4.2 (2470) | 4.0 (2530) | - | - | - |

| | |
|---|---|
| **VSWR** | ≤ 13.5 |
| **Return Loss** | ≤ -1.3 dB |
| **Peak Gain** | ≤ 4.2 dBi |

## 1.2. Supported Bands

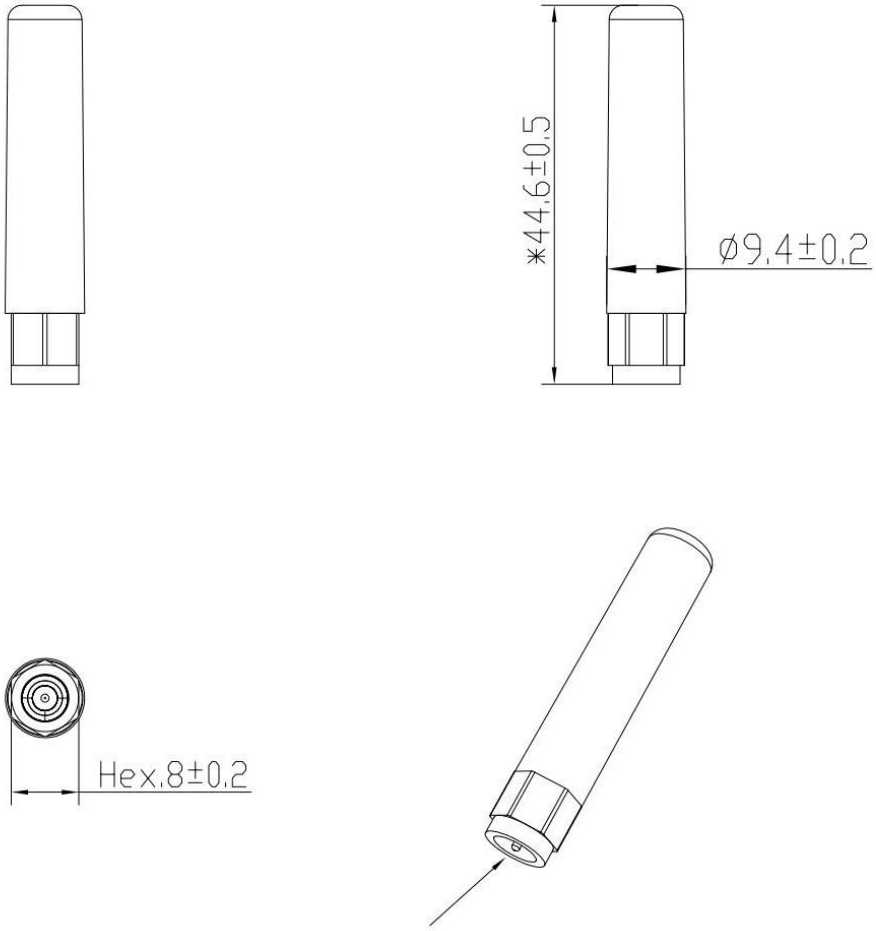
| 5G NR / LTE / LTE-Advanced / WCDMA / HSPA / HSPA+ / GPRS / GSM / NB-IoT | | | | |
|---|---|---|---|---|
| **Band** | **Frequency (MHz)** | **Uplink (MHz)** | **Downlink (MHz)** | **Covered** |
| **1** | 2100 | 1920–1980 | 2110–2170 | √ |
| **2** | 1900 | 1850–1910 | 1930–1990 | √ |
| **3** | 1800 | 1710–1785 | 1805–1880 | √ |
| **4** | 1700 | 1710–1755 | 2110–2155 | √ |
| **5** | 850 | 824–849 | 869–894 | √ |
| **7** | 2600 | 2500–2570 | 2620–2690 | √ |
| **8** | 900 | 880–915 | 925–960 | √ |
| **9** | 1800 | 1749.9–1784.9 | 1844.9–1879.9 | √ |
| **11** | 1500 | 1427.9–1447.9 | 1475.9–1495.9 | - |
| **12** | 700 | 699–716 | 729–746 | √ |
| **13** | 700 | 777–787 | 746–756 | √ |
| **14** | 700 | 788–798 | 758–768 | √ |
| **17** | 700 | 704–716 | 734–746 | √ |
| **18** | 850 | 815–830 | 860–875 | √ |
| **19** | 850 | 830–845 | 875–890 | √ |
| **20** | 800 | 832–862 | 791–821 | √ |
| **21** | 1500 | 1447.9–1462.9 | 1495.9–1510.9 | - |
| **22** | 3500 | 3410–3490 | 3510–3590 | - |
| **23** | 2100 | 2000–2020 | 2180–2200 | √ |
| **24** | 1600 | 1626.5–1660.5 | 1525–1559 | - |
| **25** | 1900 | 1850–1915 | 1930–1995 | √ |
| **26** | 850 | 814–849 | 859–894 | √ |

| | | | | |
|---|---|---|---|---|
| **28** | 700 | 703–748 | 758–803 | √ |
| **31** | 450 | 452.5–457.5 | 462.5–467.5 | - |
| **34** | 2100 | 2010–2025 | | √ |
| **38** | 2600 | 2570–2620 | | √ |
| **39** | 1900 | 1880–1920 | | √ |
| **40** | 2300 | 2300–2400 | | √ |
| **41** | 2500 | 2496–2690 | | √ |
| **42** | 3500 | 3400–3600 | | - |
| **48** | 3500 | 3550–3700 | | - |
| **66** | 1700 | 1710–1780 | 2110–2200 | √ |
| **71** | 600 | 663–698 | 617–652 | - |
| **74** | 1500 | 1427–1470 | 1475–1518 | - |
| **77** | 3500 | 3300–4200 | | - |
| **78** | 3500 | 3300–3800 | | - |
| **79** | 4500 | 4400–5000 | | - |

## 1.3. Mechanical & Environmental

| | |
|---|---|
| **Mechanical** | |
| **Antenna Dimensions** | Φ 9.4 mm × 44.6 mm |
| **Material & Color** | TPE & Black |
| **Connector Type** | SMA Male (Ni plating) |
| **Mounting Type** | Terminal |
| **Weight** | Typ. 4.08 g |
| **Environmental** | |
| **Operation Temperature** | -40 °C to +85 °C |
| **Storage Temperature** | -40 °C to +85 °C |
| **RoHS Compliant** | Yes |

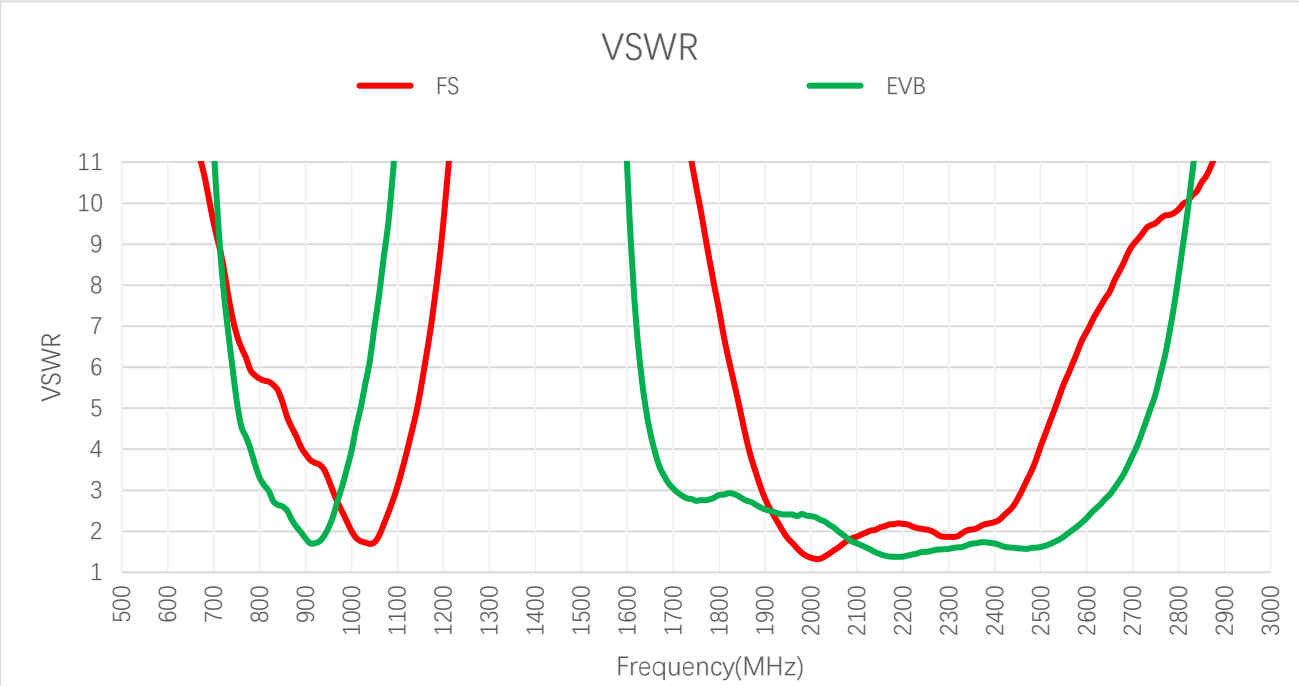
# 2 Drawing


*44.6±0.5
Ø9.4±0.2
Hex.8±0.2
SMA Male(Ni plating)
Inside screw inside pin

# 3 Detailed Performance

## 3.1. S-Parameter Test

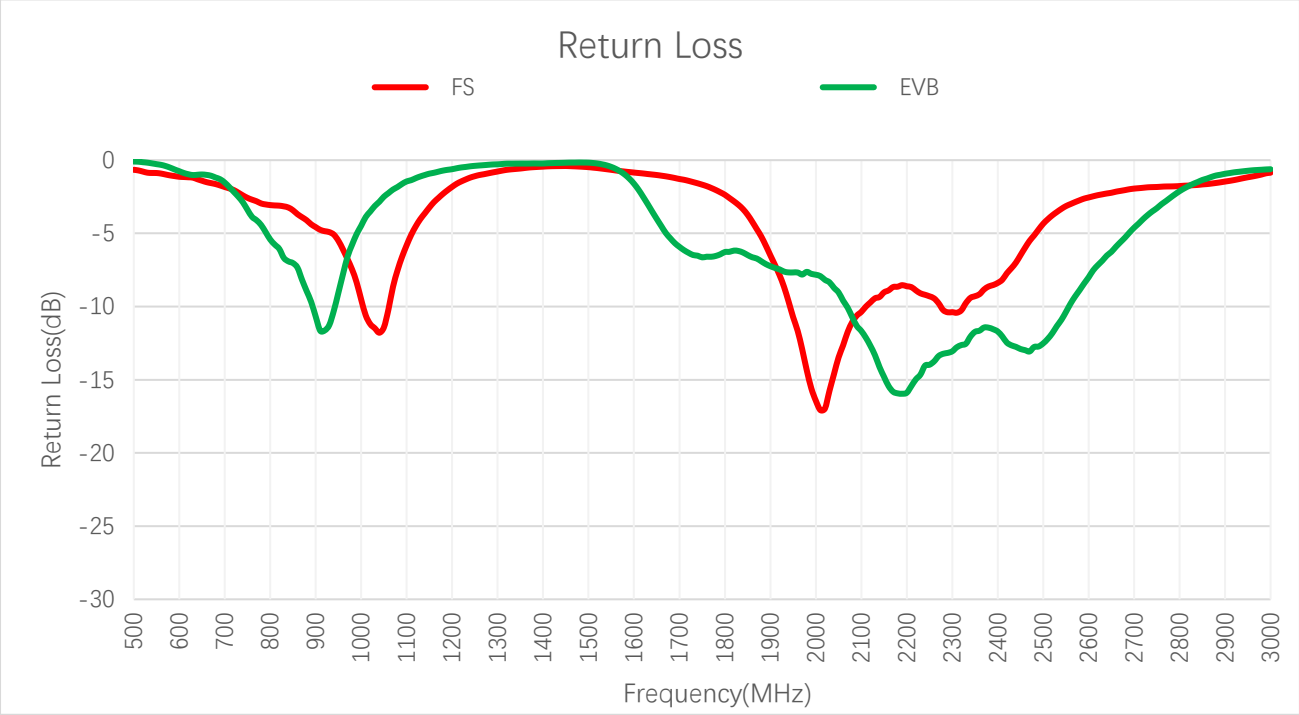
### 3.1.1. VSWR




**VSWR**

| Frequency (MHz) | 600 | 630 | 710 | 830 | 900 | 960 | 1440 | 1710 | 1740 | 1880 |
|---|---|---|---|---|---|---|---|---|---|---|
| FS | - | - | 9.0 | 5.6 | 3.9 | 3.0 | - | 13.0 | 11.0 | 3.4 |
| EVB | - | - | 9.5 | 2.7 | 1.8 | 2.4 | - | 2.9 | 2.8 | 2.6 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| FS | 1.8 | 2.0 | 2.0 | 2.8 | 6.9 | 8.8 | - | - | - | - |
| EVB | 2.4 | 1.5 | 1.7 | 1.6 | 2.3 | 3.6 | - | - | - | - |

### 3.1.2. Return Loss

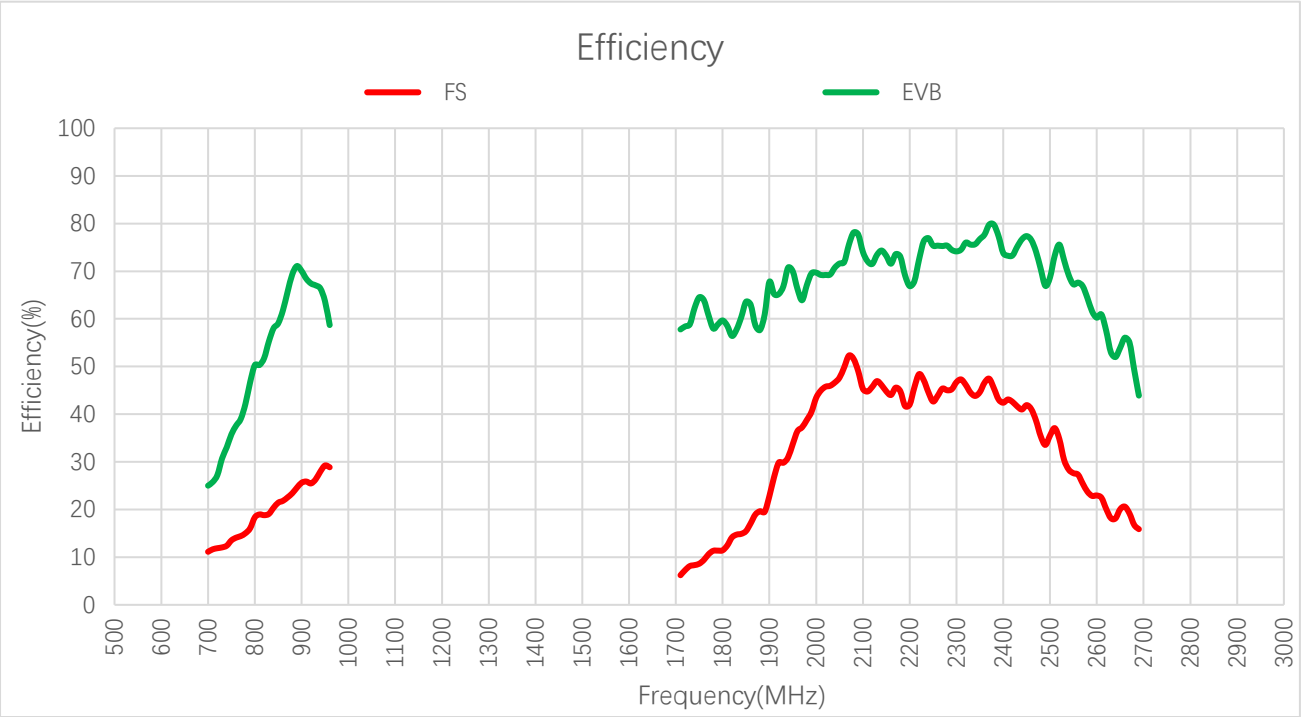



**Return Loss (dB)**

| Frequency (MHz) | 600 | 630 | 710 | 830 | 900 | 960 | 1440 | 1710 | 1740 | 1880 |
|---|---|---|---|---|---|---|---|---|---|---|
| FS | - | - | -1.9 | -3.2 | -4.6 | -6.1 | - | -1.3 | -1.6 | -5.2 |
| EVB | - | - | -1.8 | -6.7 | -10.7 | -7.8 | - | -6.1 | -6.5 | -6.9 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| FS | -10.7 | -9.3 | -9.3 | -6.5 | -2.5 | -2.0 | - | - | - | - |
| EVB | -7.7 | -14.1 | -11.7 | -12.9 | -8.0 | -4.9 | - | - | - | - |

## 3.2. Radiation Performance Test

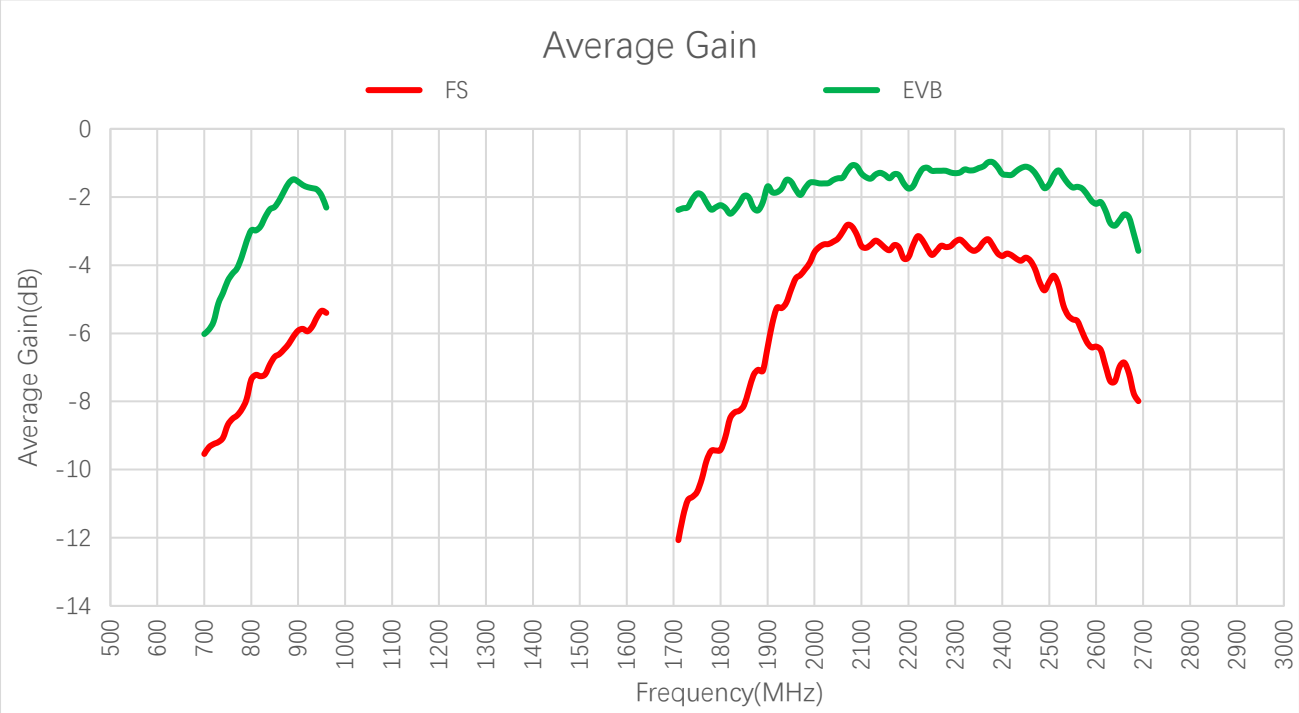
### 3.2.1. Efficiency




**Efficiency (%)**

| Frequency (MHz) | 600 | 630 | 710 | 830 | 900 | 960 | 1440 | 1710 | 1740 | 1880 |
|---|---|---|---|---|---|---|---|---|---|---|
| FS | - | - | 11.7 | 19.1 | 25.6 | 28.9 | - | 6.2 | 8.3 | 19.6 |
| EVB | - | - | 25.8 | 55.3 | 70.1 | 58.7 | - | 57.8 | 62.2 | 57.7 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| FS | 33.7 | 46.1 | 44.6 | 41.9 | 22.9 | 15.9 | - | - | - | - |
| EVB | 70.0 | 74.3 | 76.7 | 77.4 | 60.3 | 43.9 | - | - | - | - |

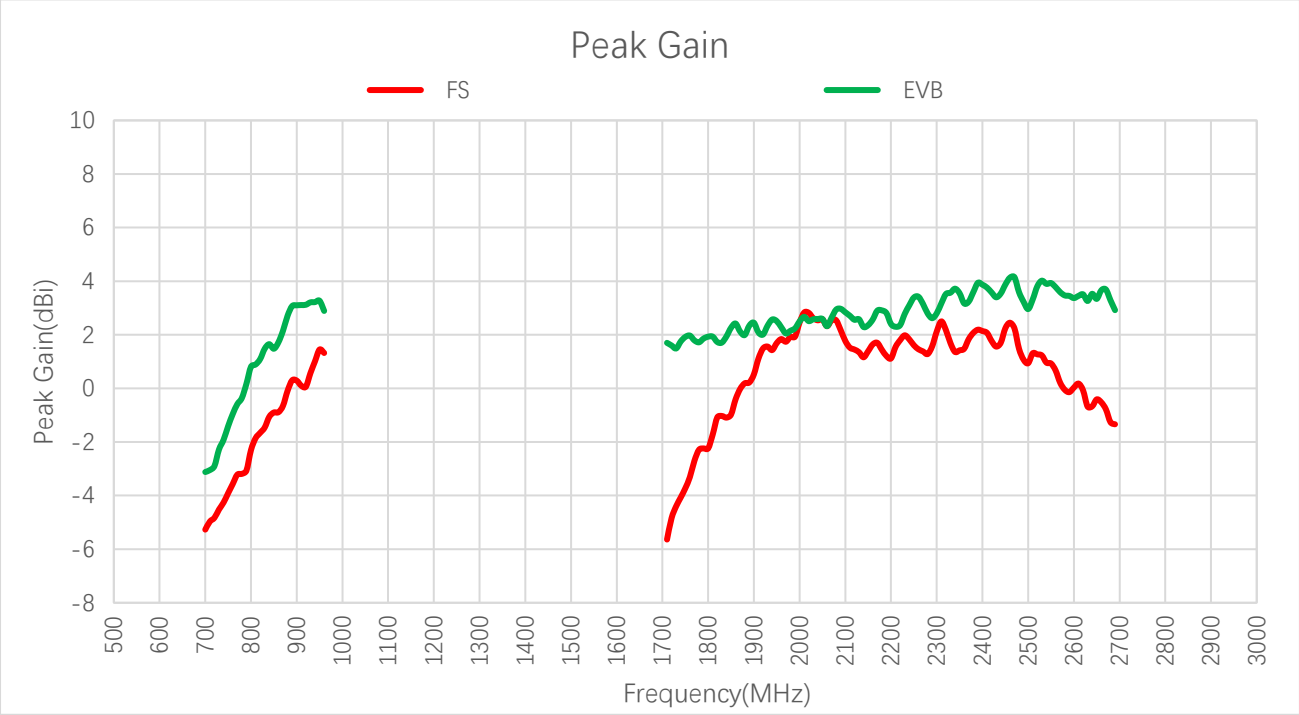
### 3.2.2. Average Gain




**Average Gain (dB)**

| Frequency (MHz) | 600 | 630 | 710 | 830 | 900 | 960 | 1440 | 1710 | 1740 | 1880 |
|---|---|---|---|---|---|---|---|---|---|---|
| FS | - | - | -9.3 | -7.2 | -5.9 | -5.4 | - | -12.1 | -10.8 | -7.1 |
| EVB | - | - | -5.9 | -2.6 | -1.6 | -2.3 | - | -2.4 | -2.1 | -2.4 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| FS | -4.7 | -3.4 | -3.5 | -3.8 | -6.4 | -8.0 | - | - | - | - |
| EVB | -1.6 | -1.3 | -1.2 | -1.1 | -2.2 | -3.6 | - | - | - | - |

### 3.2.3. Peak Gain



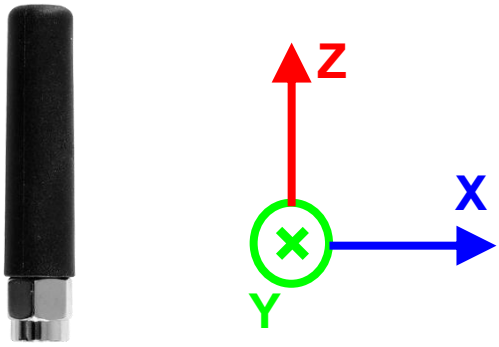

**Peak Gain (dBi)**

| Frequency (MHz) | 600 | 630 | 710 | 830 | 900 | 960 | 1440 | 1710 | 1740 | 1880 |
|---|---|---|---|---|---|---|---|---|---|---|
| FS | - | - | -5.0 | -1.5 | 0.3 | 1.3 | - | -5.6 | -4.1 | 0.2 |
| EVB | - | - | -3.1 | 1.5 | 3.1 | 2.9 | - | 1.7 | 1.8 | 2.0 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| FS | 1.7 | 1.2 | 1.4 | 2.2 | 0.0 | -1.3 | - | - | - | - |
| EVB | 2.5 | 2.3 | 3.6 | 3.9 | 3.4 | 2.9 | - | - | - | - |

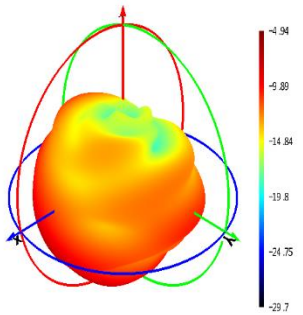
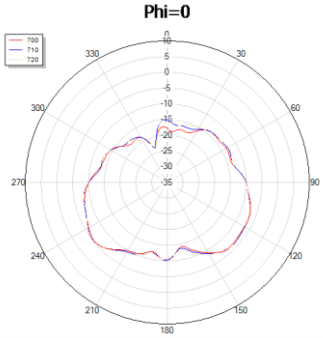
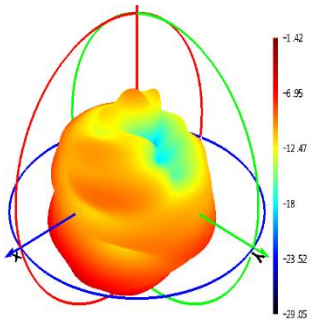
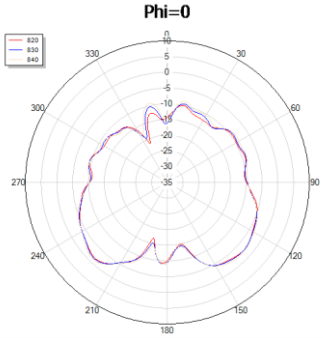
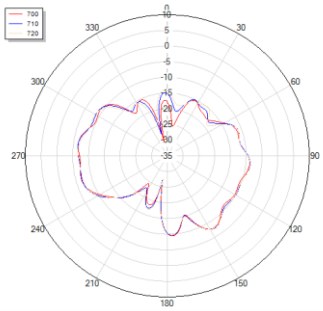
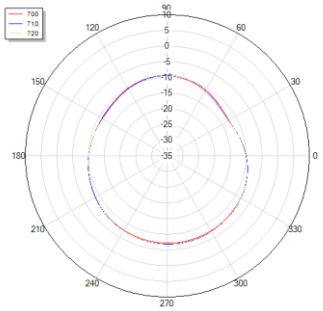
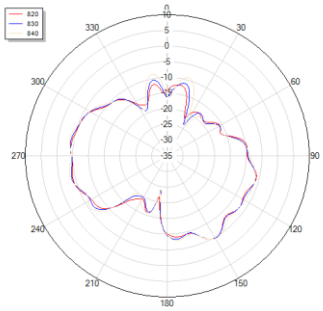
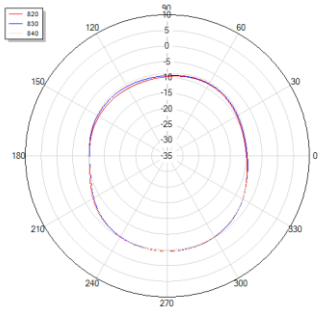
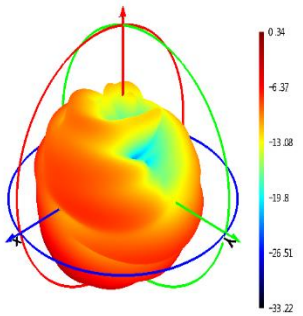
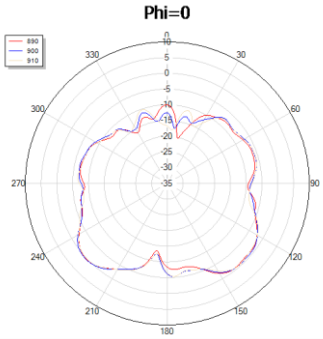
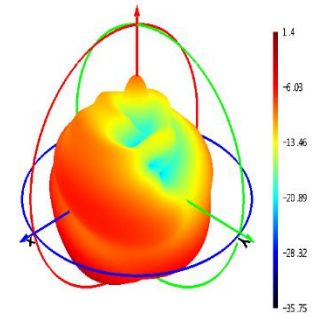
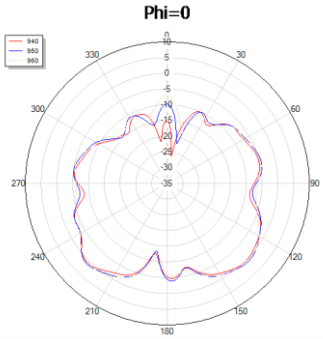
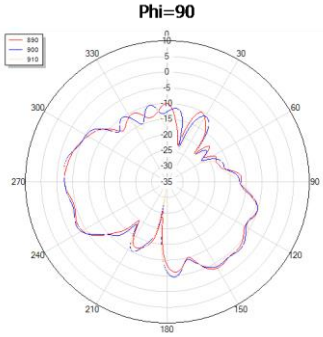
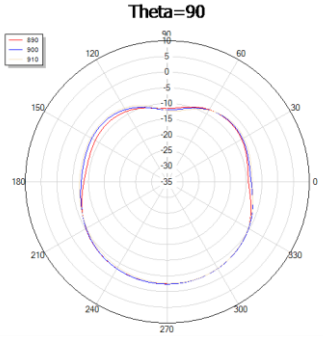
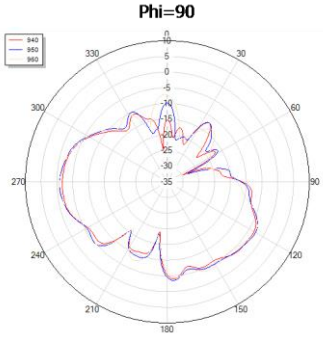
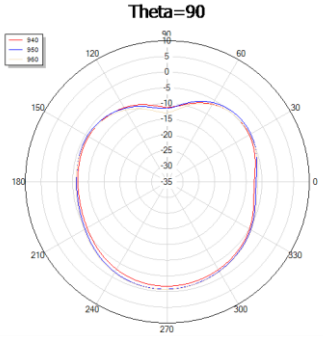
### 3.2.4. 3D & 2D Radiation Pattern

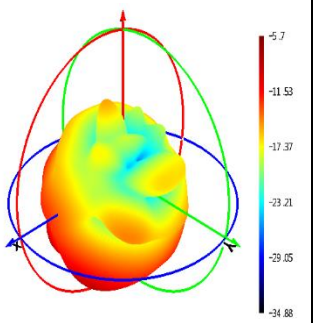
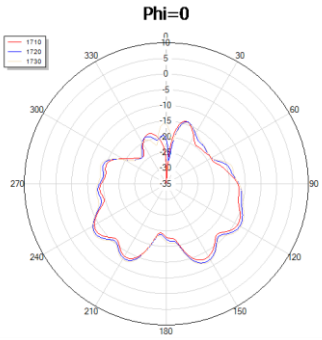
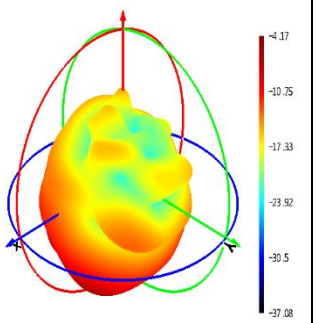
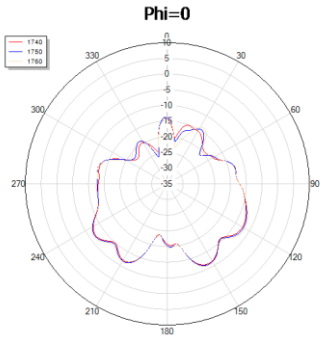
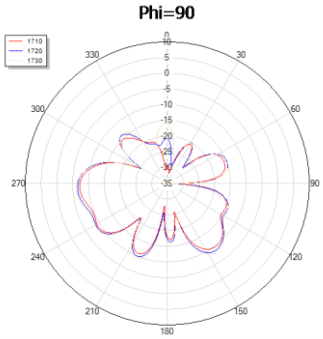
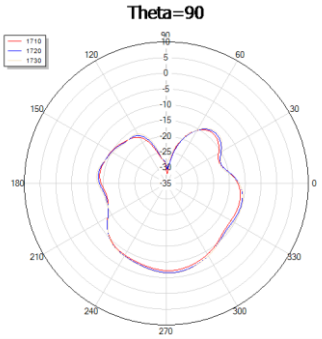
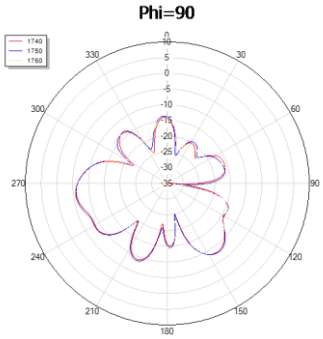
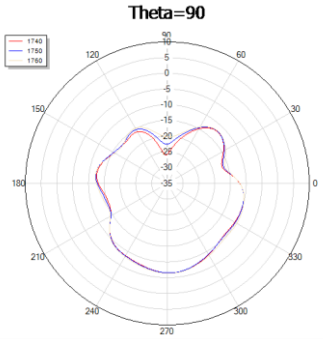
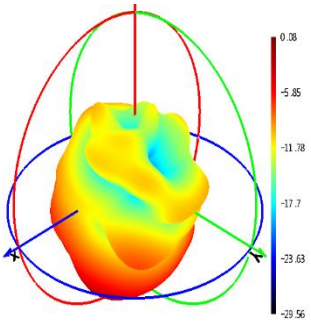
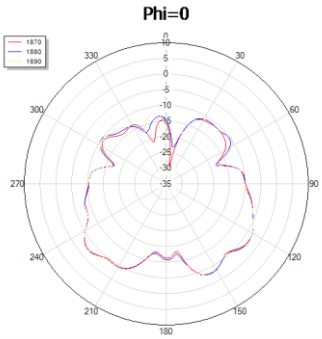
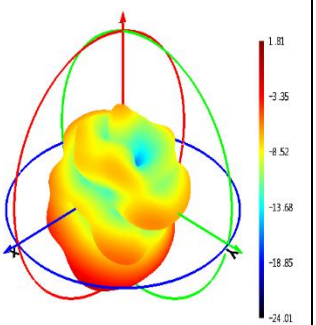
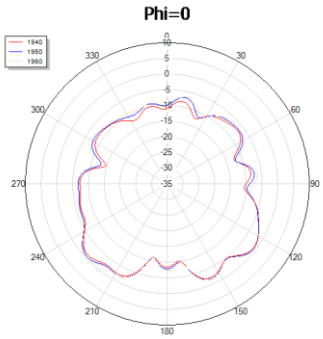
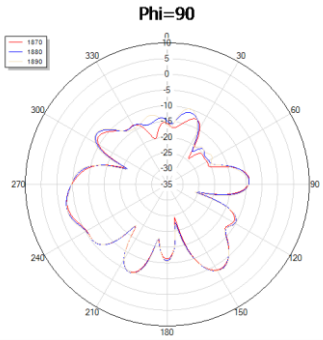
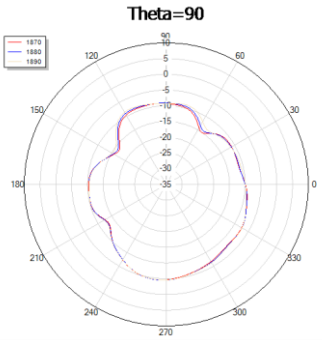
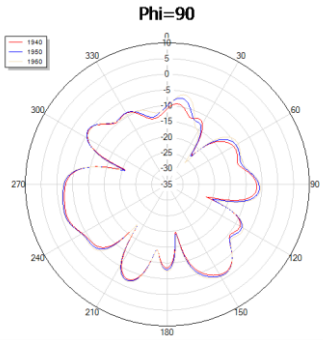
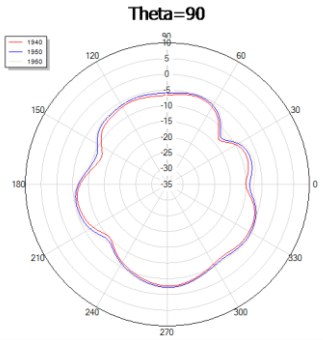
#### 3.2.4.1. Test Condition: Free Space

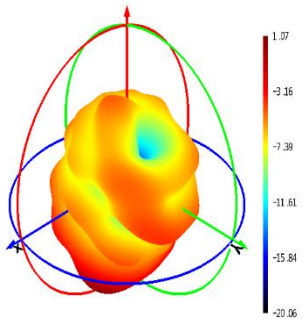
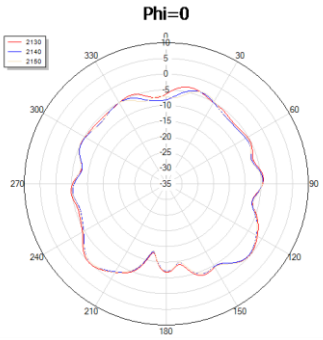
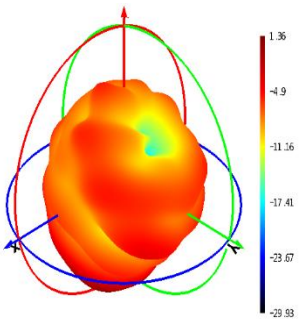
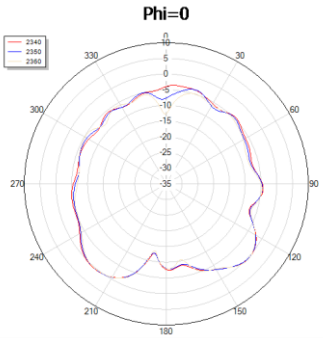
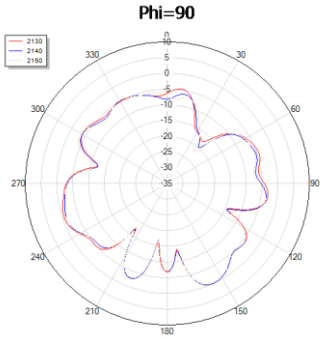
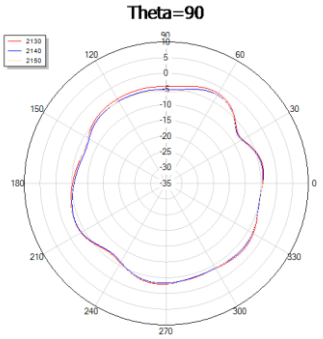
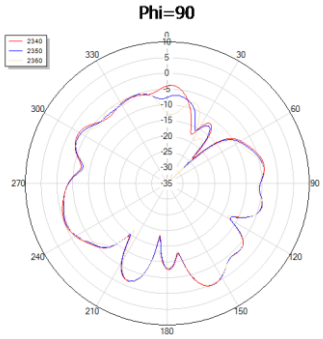
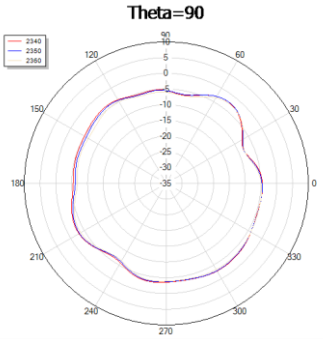
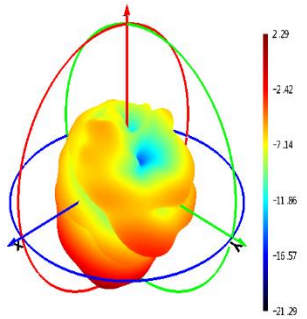
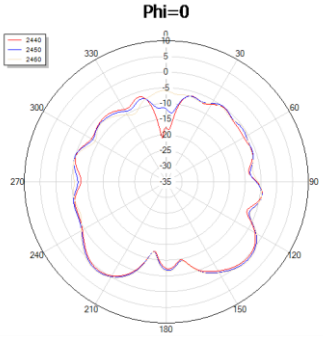
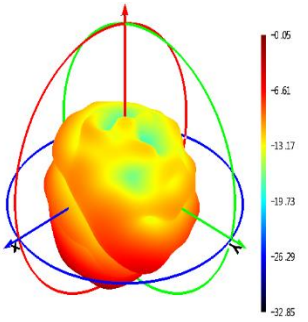
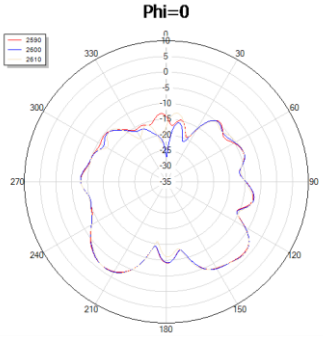
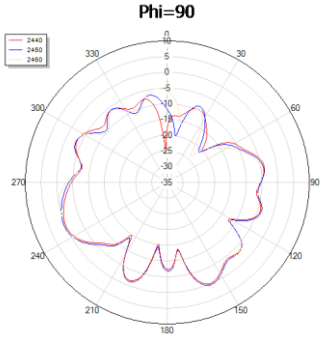
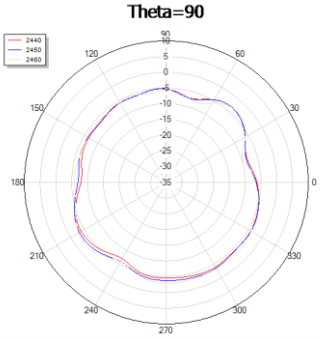
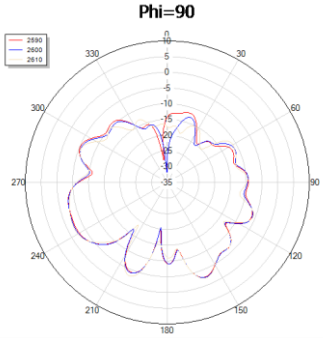
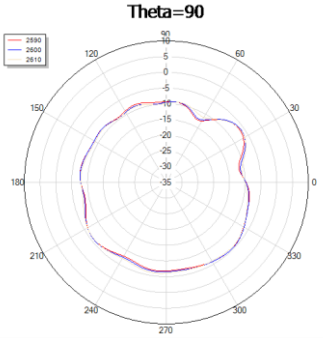
- Test Chamber: HF-SY-1

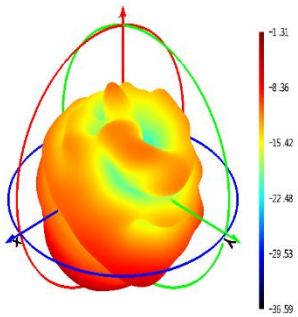
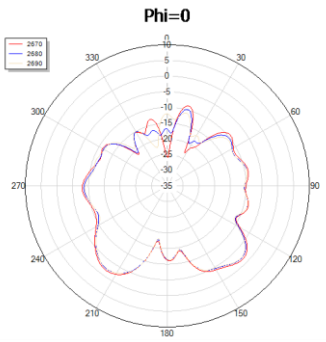
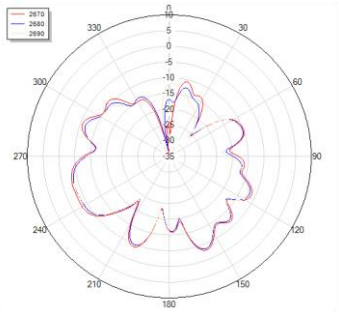
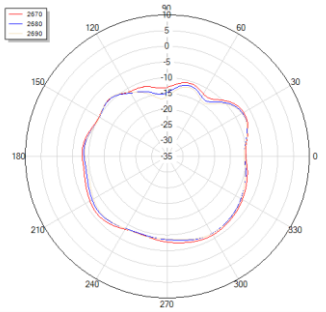


- **FS**

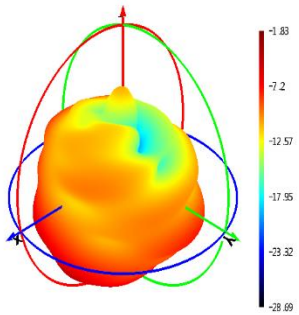
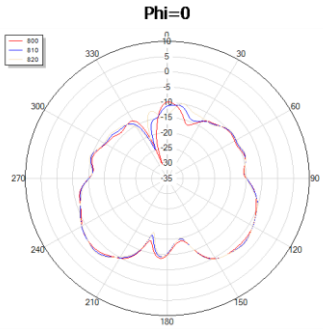
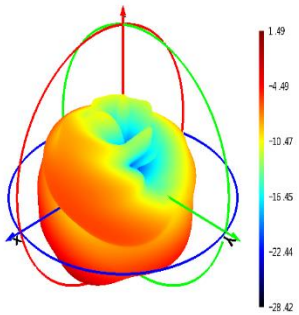
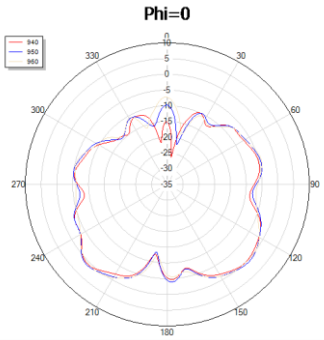
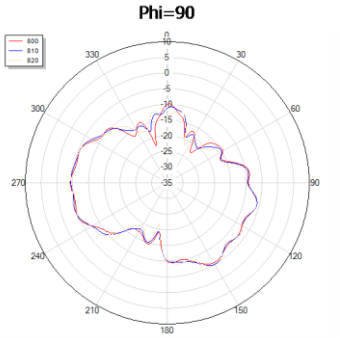
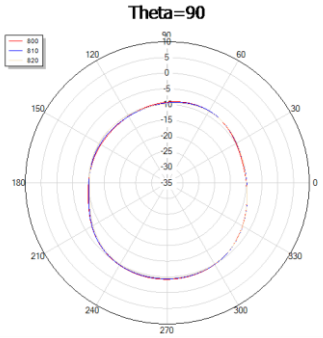
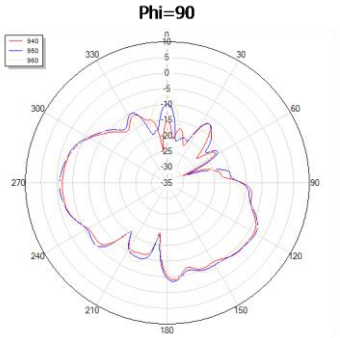
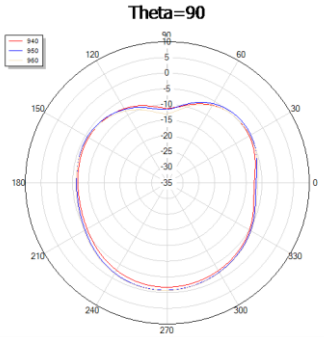
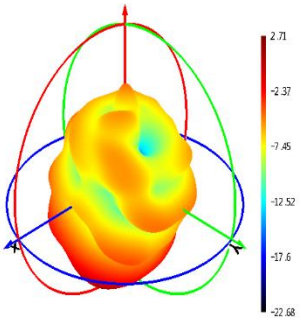
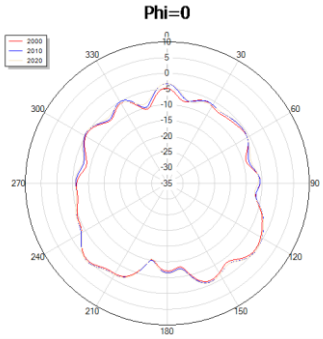
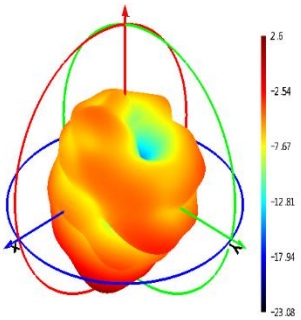
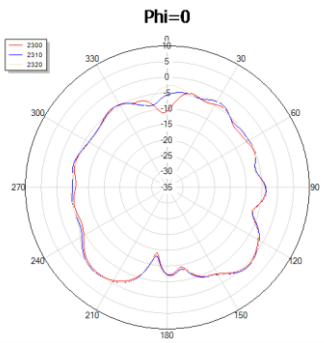
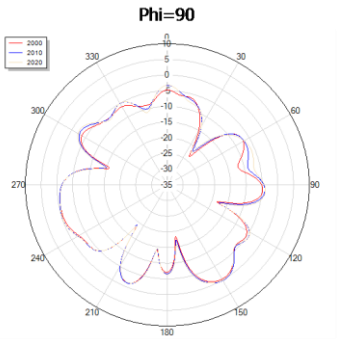
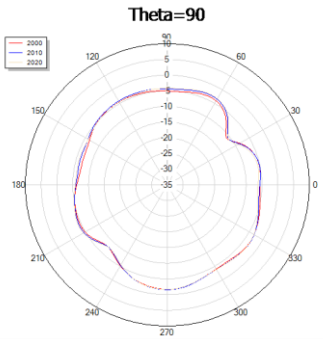
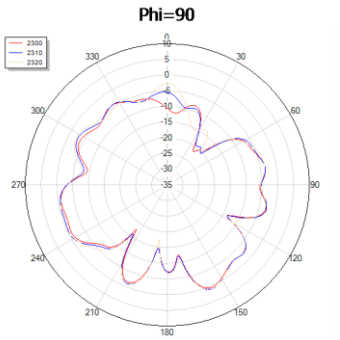
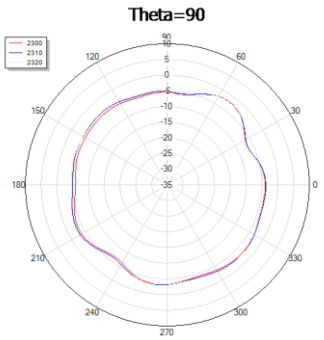
| 710 MHz | | 830 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **900 MHz** | | **960 MHz** | |
|  |  |  |  |
|  |  |  |  |

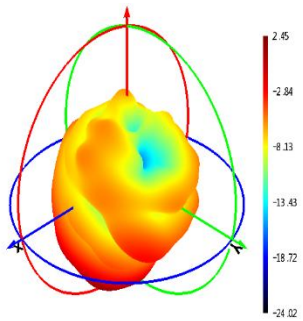
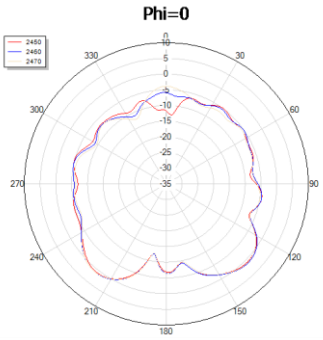
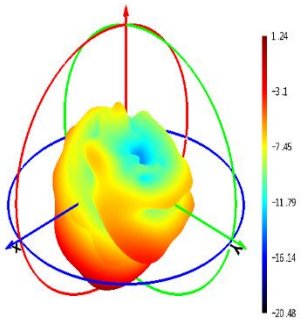
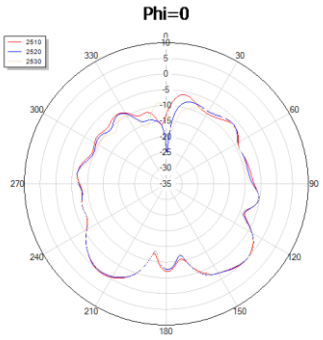
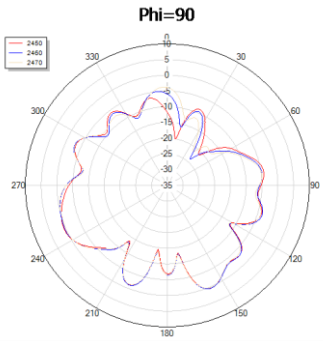
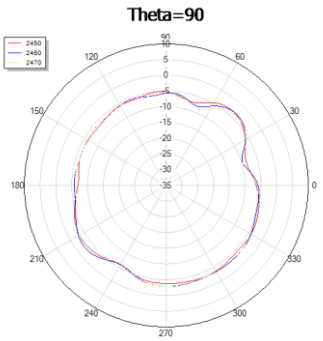
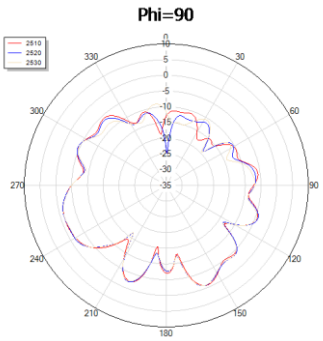
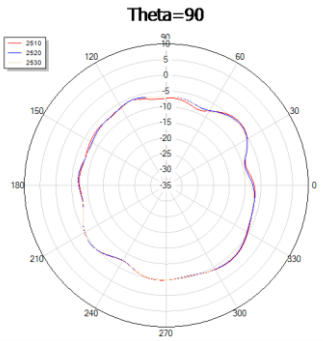
| 1710 MHz | | 1740 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **1880 MHz** | | **1950 MHz** | |
|  |  |  |  |
|  |  |  |  |

| 2140 MHz | | 2350 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **2450 MHz** | | **2600 MHz** | |
|  |  |  |  |
|  |  |  |  |

| 2690 MHz | |
|---|---|
|  |  |
|  |  |

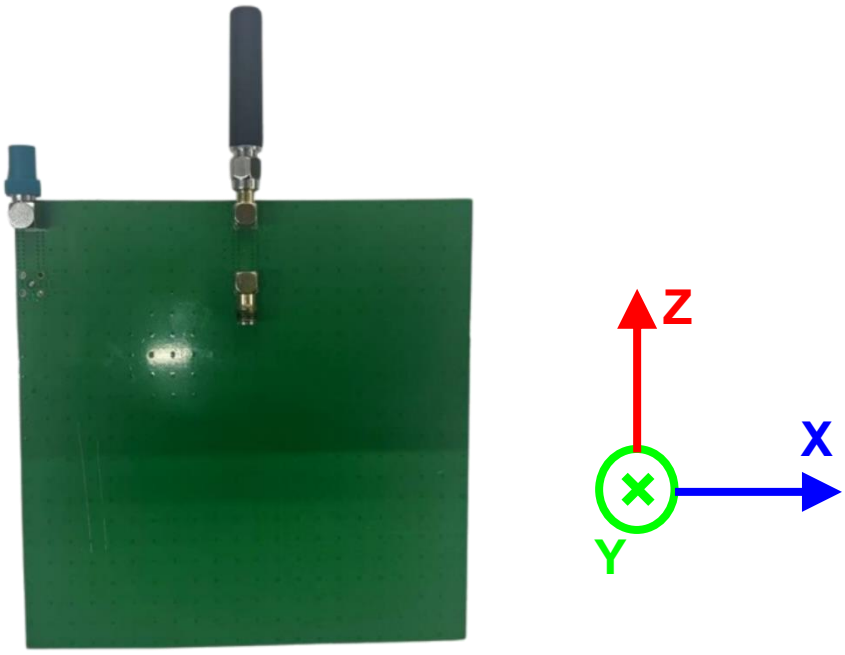
- **Max Peak Gain**

| 810 MHz | | 950 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **2010 MHz** | | **2310 MHz** | |
|  |  |  |  |
|  |  |  |  |

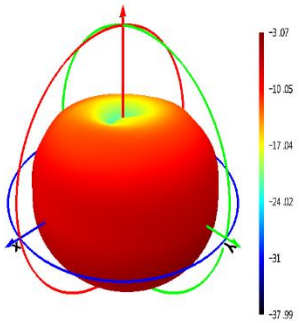
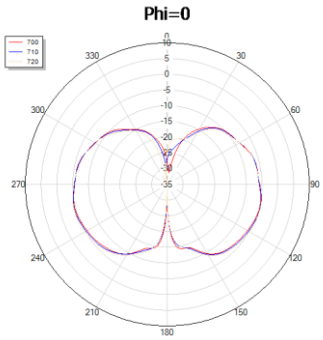
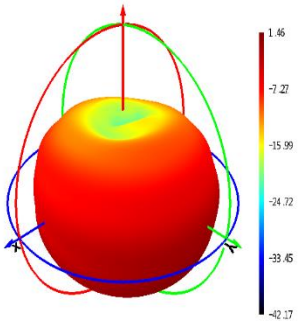
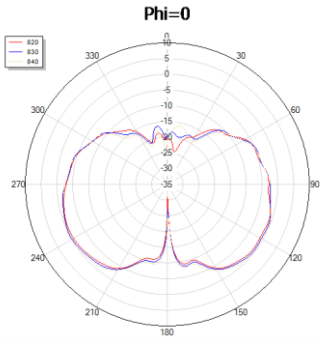
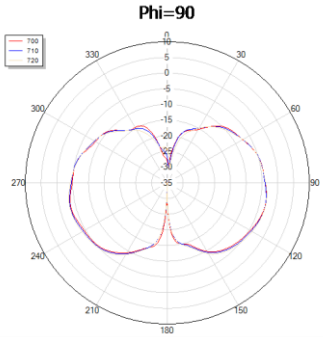
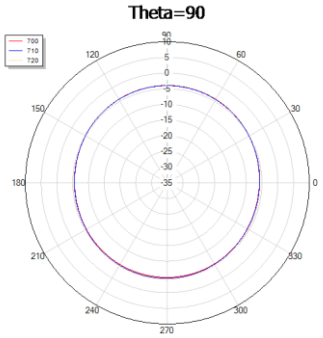
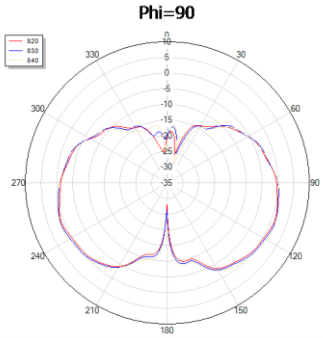
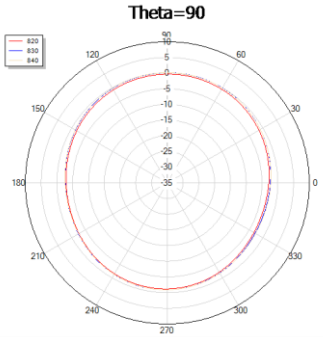
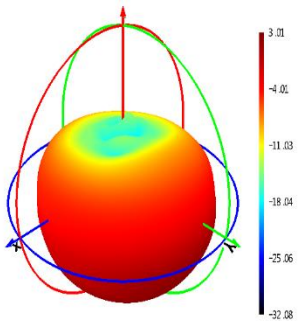
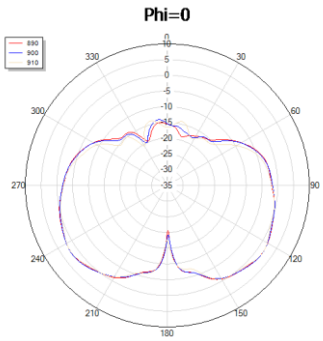
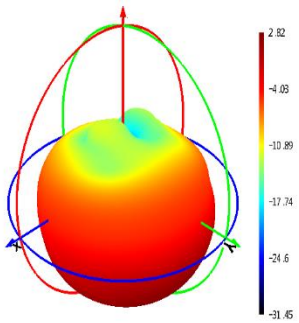
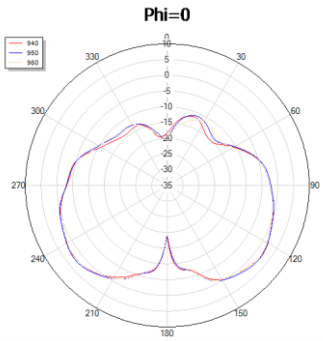
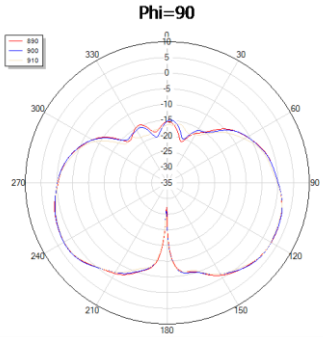
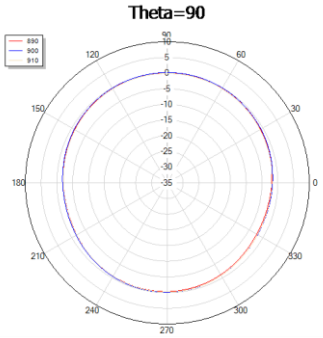
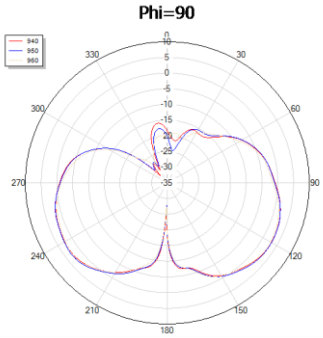
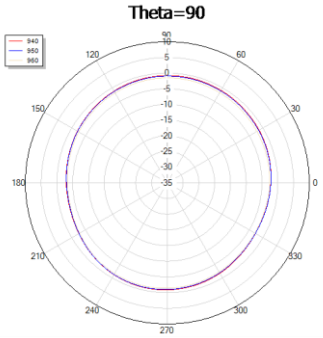
| 2460 MHz | | 2520 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |

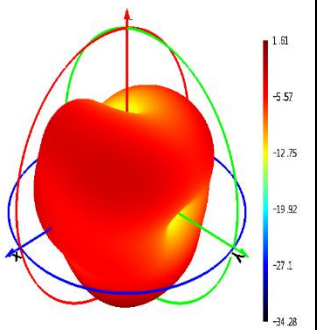
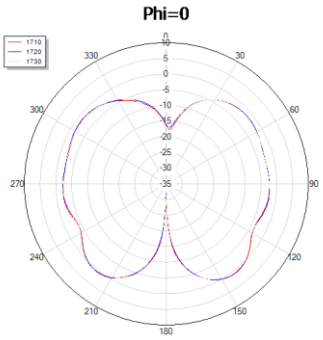
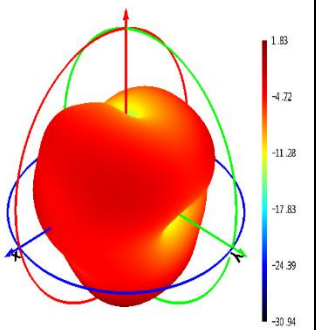
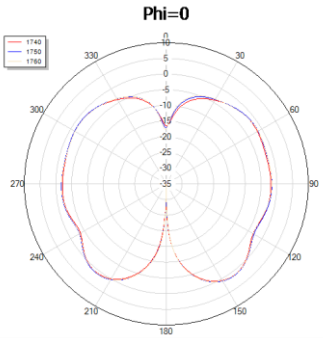
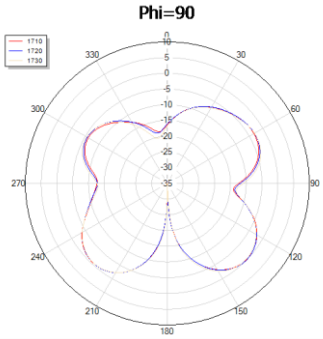
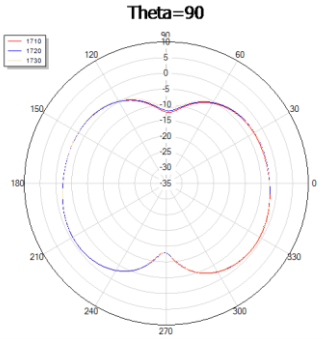
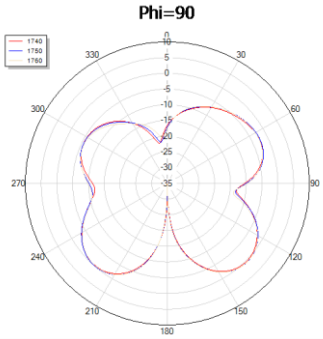
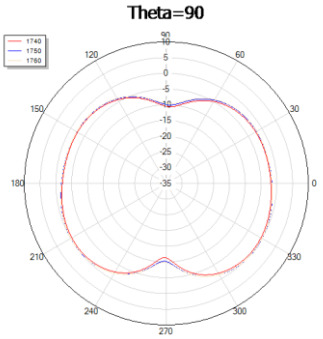
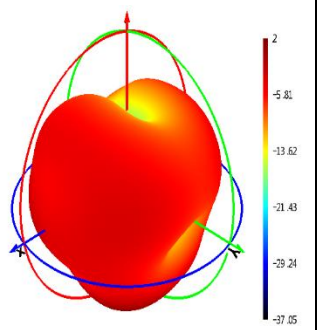
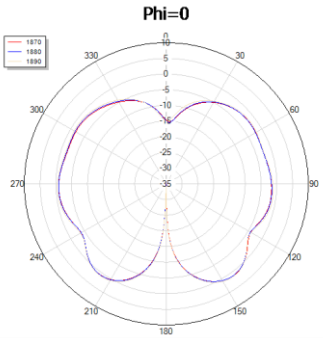
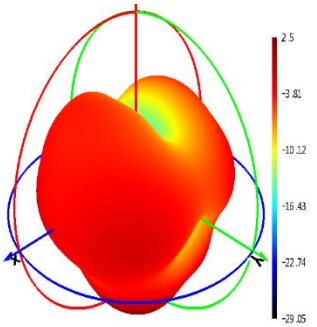
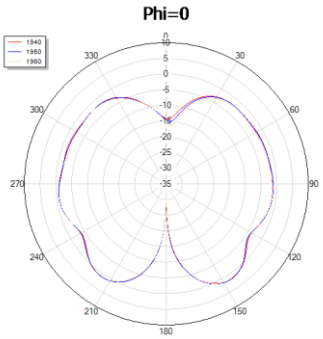
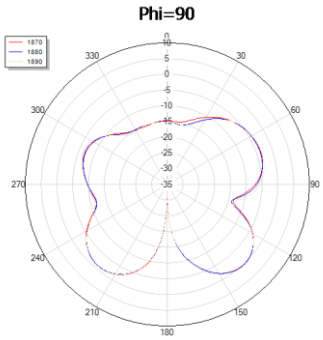
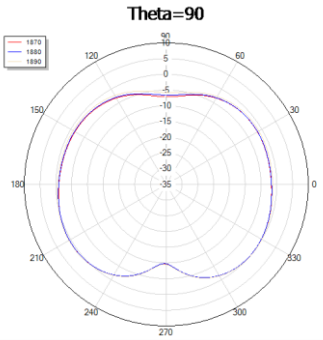
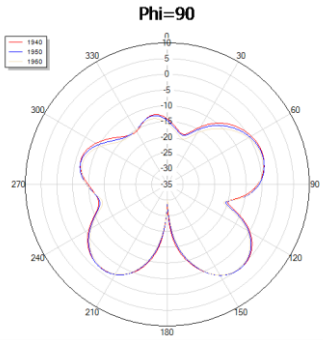
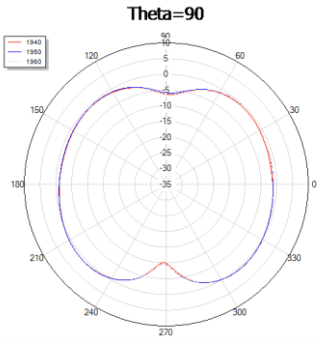
#### 3.2.4.2. Test Condition: On 130 mm × 130 mm EVB

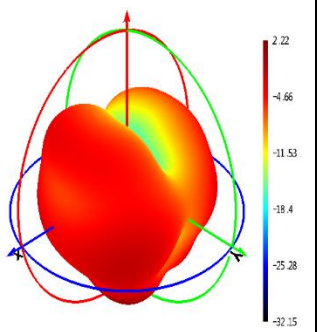
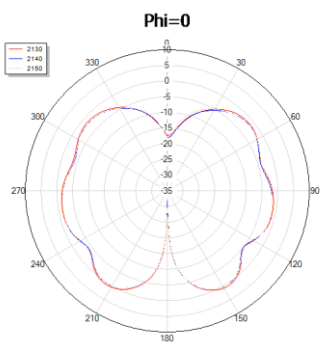
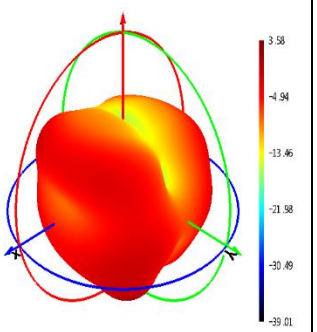
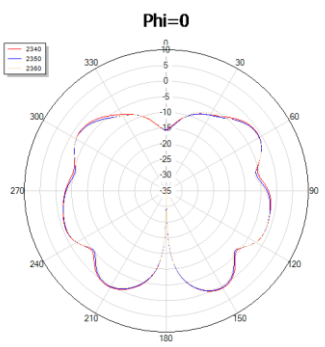
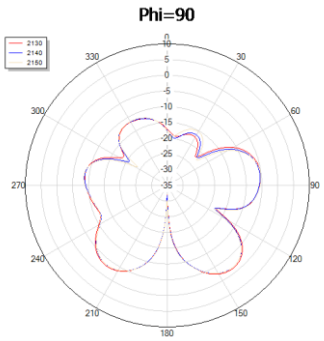
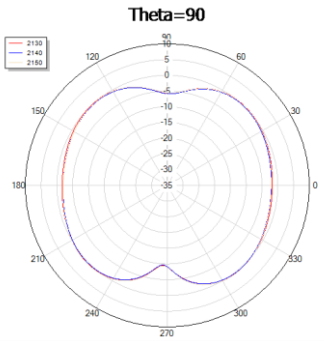
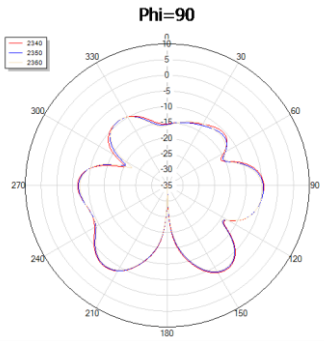
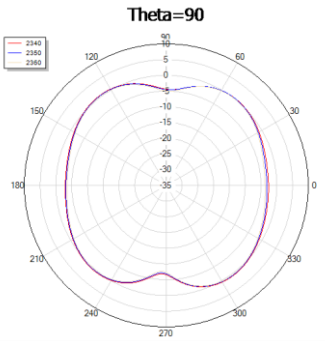
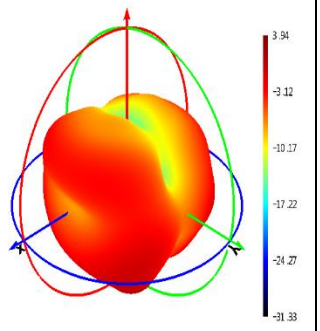
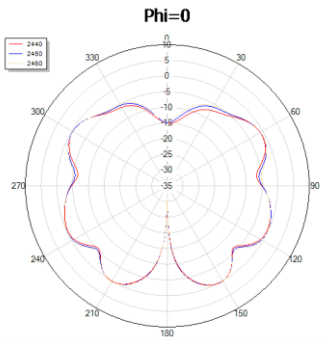
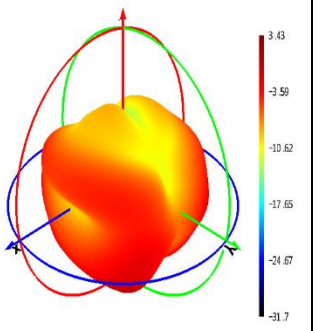
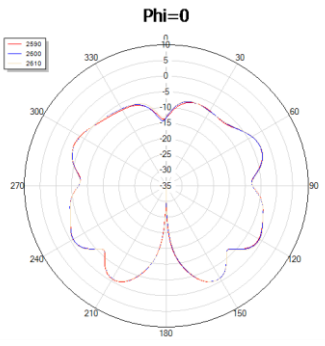
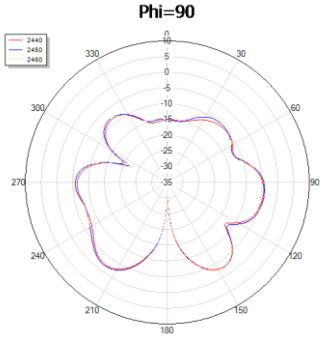
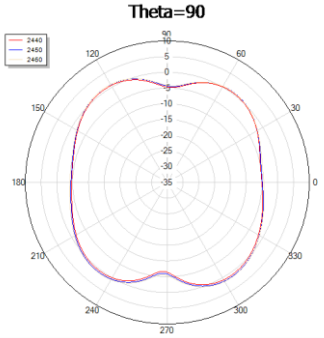
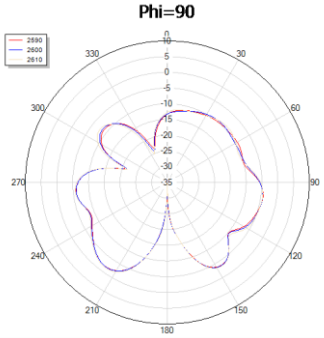
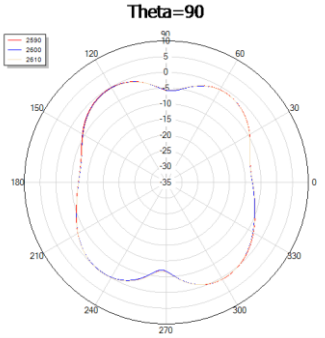
- Test Chamber: HF-SY-1

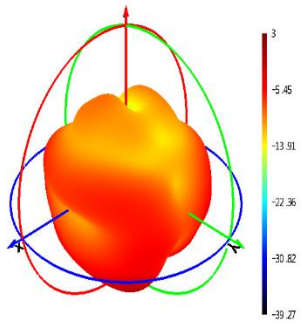
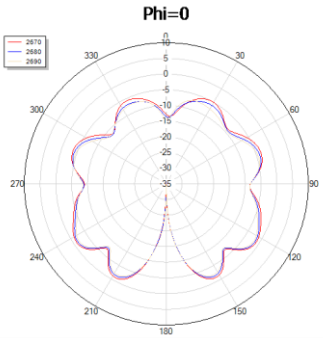
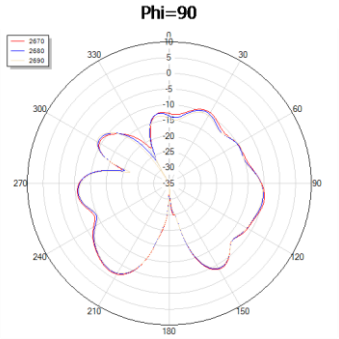
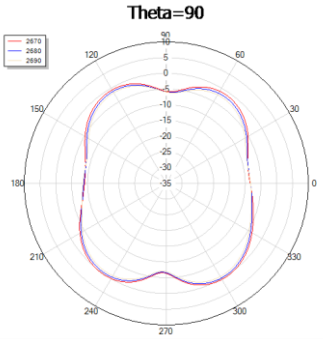


- **EVB**

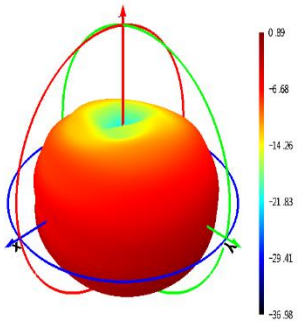
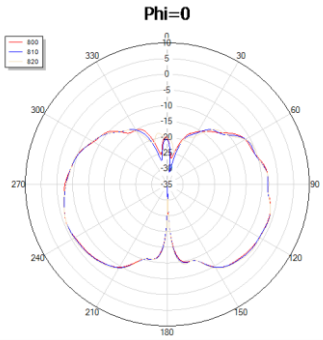
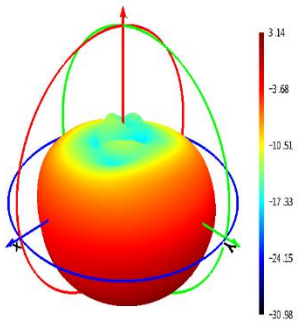
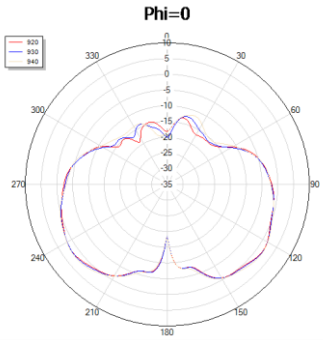
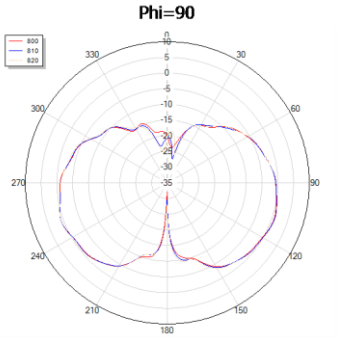
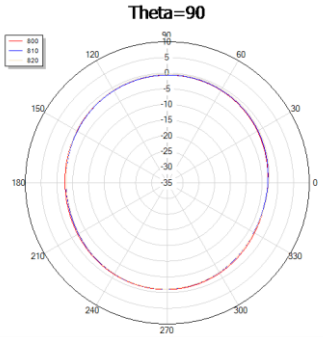
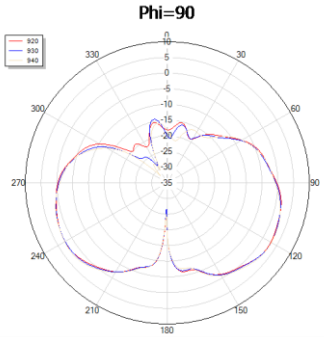
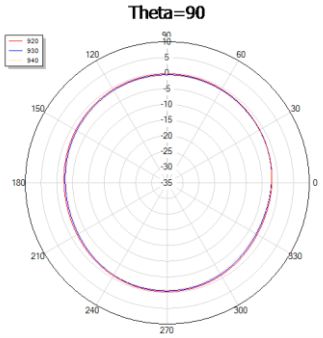
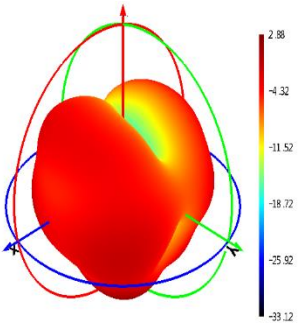
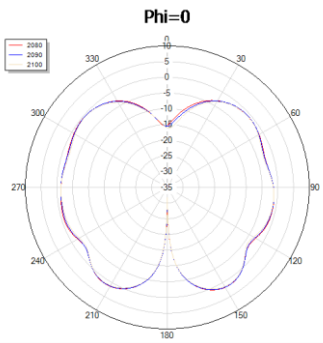
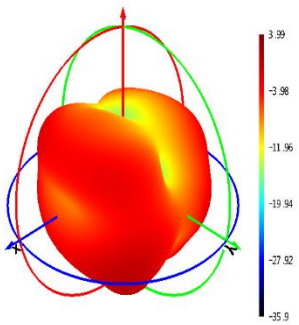
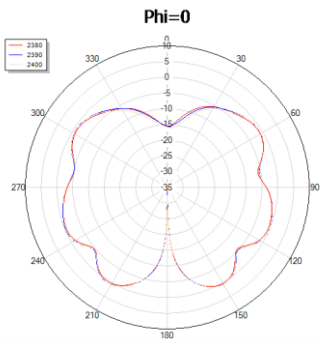
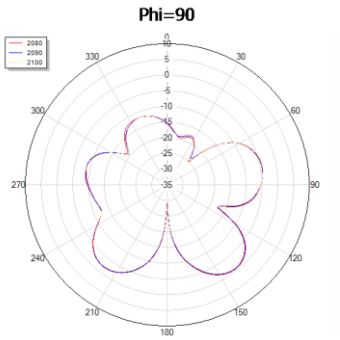
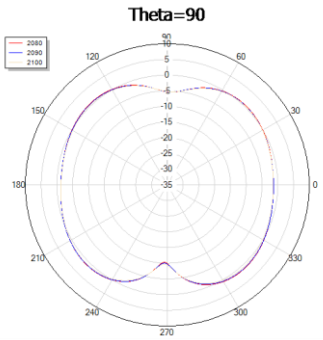
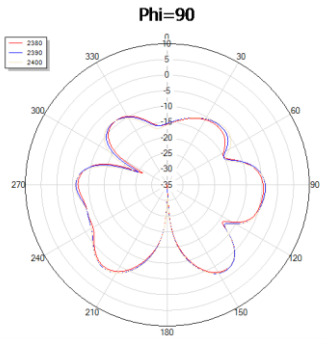
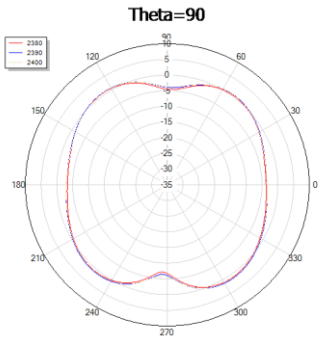
| 710 MHz | | 830 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **900 MHz** | | **960 MHz** | |
|  |  |  |  |
|  |  |  |  |

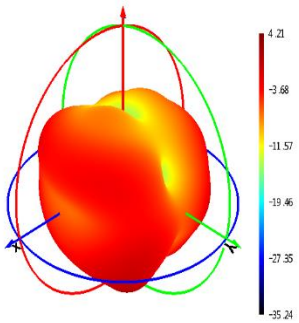
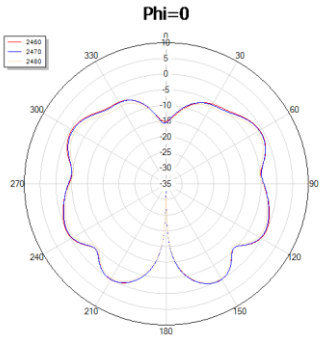
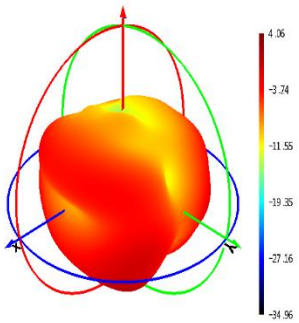
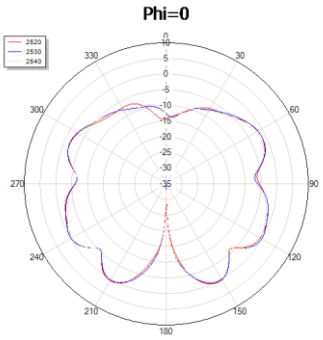
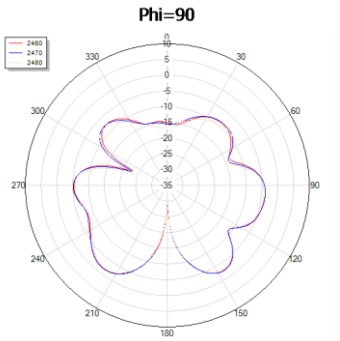
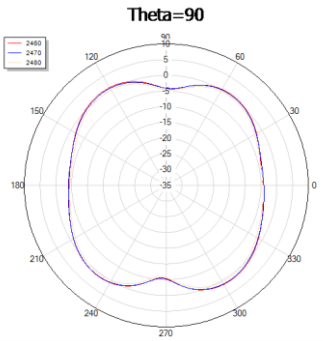
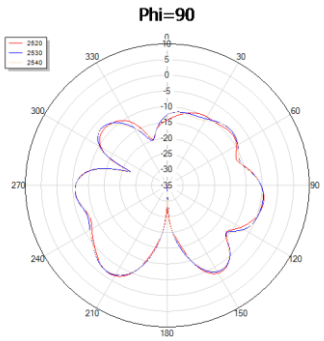
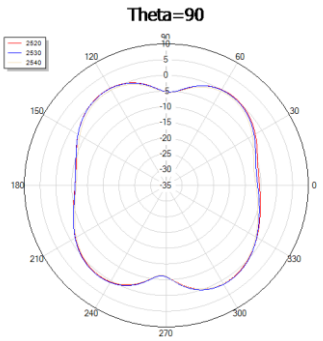
| 1710 MHz | | 1740 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **1880 MHz** | | **1950 MHz** | |
|  |  |  |  |
|  |  |  |  |

| 2140 MHz | | 2350 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **2450 MHz** | | **2600 MHz** | |
|  |  |  |  |
|  |  |  |  |

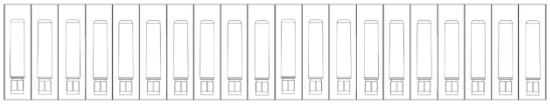
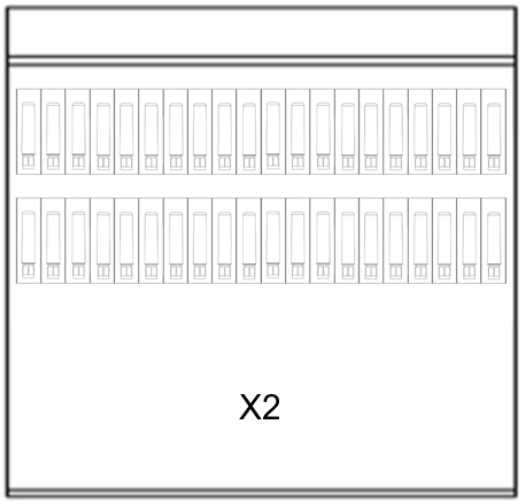
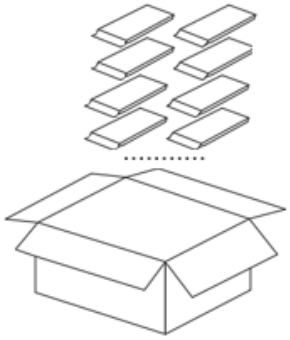
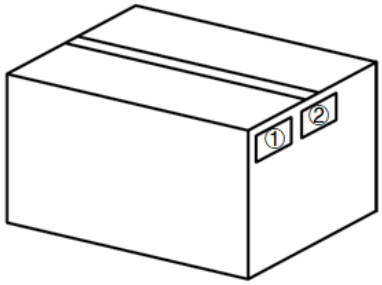
| 2690 MHz | |
|---|---|
|  3<br>-5.45<br>-13.91<br>-22.36<br>-30.82<br>-39.27 |  Phi=0<br>2670<br>2680<br>2690 |
|  Phi=90<br>2670<br>2680<br>2690 |  Theta=90<br>2670<br>2680<br>2690 |

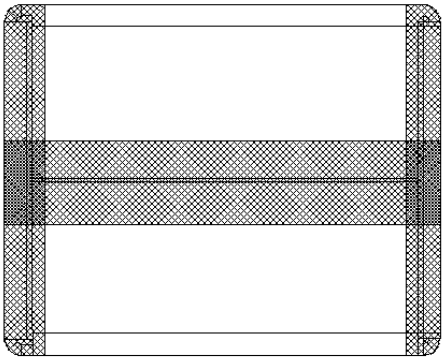
- **Max Peak Gain**

| 810 MHz | | 930 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| **2090 MHz** | | **2390 MHz** | |
|  |  |  |  |
|  |  |  |  |

| 2470 MHz | | 2530 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |

# 4 Packaging

| Step | Packaging Picture / 2D Picture | Description |
|---|---|---|
| 1 |  | 20 antenna products in a one-piece bag.<br>(20 Antennas / One-piece Bag) |
| 2 |  | 40 antenna products in a PE bag.<br>(40 Antennas / PE Bag) |
| 3 |  | (48 PE Bags / Carton Box)<br>(1920 Antennas / Carton Box)<br>Estimated quantity<br>Products that cannot fill the entire carton box are packed in a suitable size carton box.<br>Carton Size:<br>L × W × H = 470 × 310 × 310 mm |
| 4 |  | **Position for Attaching Labels**<br>① Carton Label<br>② Quality Label |

| | | |
|---|---|---|
| 5 |  | **Sealing Cartons**<br>H-shaped sealing cartons |
| Note | The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging. | |

# Contact Us

**At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:**

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Email: info@quectel.com

**Or our local offices. For more information, please visit:**
https://www.quectel.com/contact/.

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# Revision History

| Version | Date | Author | Note |
|---|---|---|---|
| - | 2026-05-25 | Simon Sheng/ Toby Wang/ Caius Zhang/ Rainey Liao | Creation of the document |
| 1.0 | 2026-05-25 | Simon Sheng/ Toby Wang/ Caius Zhang/ Rainey Liao | First official release |



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